Computation of Symmetric Homoclinic Orbits in the Restricted Three Body Problem

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Introduction

In this document, we numerically compute symmetric homoclinic orbits associated with the equilibrium point L_3 in the restricted three body problem. We begin by introducing the planar restricted three body problem, and then we explain how to compute symmetric homoclinic orbits based on the unstable invariant curve associated with L_3. We numerically determine the values of the mass parameter μ for which this procedure is possible by computing the x-component of the velocity of the unstable curve at its first intersection with the x-axis, as a function of μ. Finally, we include the *SageMath* code used for the numerical computations and for generating the plots.

Part 1: The Restricted Three Body Problem

This section provides a brief introduction to the aspects of the restricted three body problem (RTBP) relevant to our objectives in this document.
The considered RTBP is based on the following assumptions:

- The three bodies considered in the problem have different masses. The primaries are the two bodies with larger mass, namely $m_1 \geq m_2$, while the third body has infinitesimal mass ($m_3 \approx 0$). The primaries exert a gravitational influence on the dynamics of the third body, but the third body does not affect the motion of the primaries.

- The primaries describe circular orbits around their common center of mass, located at the origin of a plane, in a set of coordinates named sidereal coordinates. The motion of the primaries is known.

- The third body orbit is contained in the plane of motion of the primaries (planar RTBP). The motion of the third body is unknown, and its dynamics can be described by applying Newton's law under the gravitational influence of the primaries in sidereal coordinates.

- There exists a rotating system of coordinates in which the primaries remain at rest, called the synodic coordinates. In these coordinates, the equations of motion of the third body form a system of two autonomous second-order differential equations, or equivalently, a system of four autonomous ordinary differential equations. We study the dynamics of the third body through this system of four differential equations in synodic coordinates.

Let us consider the RTBP in synodic coordinates. The positions of the primaries are $P_1 = (\mu, 0)$ (mass m_1) and $P_2 = (\mu - 1, 0)$ (mass m_2), with $\mu = \frac{m_1}{m_1+m_2} \in (0, 0.5]$ the mass parameter.

Let $(x_1, x_2) = (x_1(t), x_2(t)) = (x, y)$ denote the position of the third body in synodic coordinates as a function of time, and let $(x_3, x_4) = (x_3(t), x_4(t)) = (\frac{d}{dt}x_1(t), \frac{d}{dt}x_2(t))$ denote its velocity.
Let $r_1 = \sqrt{(x_1 - \mu)^2 + x_2^2}$ and $r_2 = \sqrt{(x_1 - \mu + 1)^2 + x_2^2}$ denote the distances from the primaries to the third body.
The equations of motion of the third body in synodic coordinates are as follows:

$$\begin{cases} x_1' = x_3, \\ x_2' = x_4, \\ x_3' = 2x_4 + \Omega_x(x_1, x_2), \\ x_4' = -2x_3 + \Omega_y(x_1, x_2), \end{cases} \tag{1}$$

where $'$ denotes $\frac{d}{dt}$, and

$$\begin{aligned}\Omega_x(x_1,x_2) &= x_1 - \frac{(1-\mu)(x_1-\mu)}{r_1^3} - \frac{\mu(x_1-\mu+1)}{r_2^3},\\ \Omega_y(x_1,x_2) &= x_2(1-\frac{1-\mu}{r_1^3}-\frac{\mu}{r_2^3}).\end{aligned} \tag{2}$$

Notably, the system (1) has the following symmetry:

If

$$\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t), x_4(t))$$

is a solution of the system, then

$$\mathbf{X}(t) = \big(X_1(t), X_2(t), X_3(t), X_4(t)\big) = (x_1(-t), -x_2(-t), -x_3(-t), x_4(-t))$$

is also a solution.
Indeed, note that:

$$\begin{aligned}X_1'(t) &= \frac{d}{dt}(x_1(-t)) = -x_1'(-t) = -x_3(-t) = X_3(t),\\ X_2'(t) &= \frac{d}{dt}(-x_2(-t)) = x_2'(-t) = x_4(-t) = X_4(t),\end{aligned}$$

and

$$\begin{aligned}X_3'(t) &= x_3'(-t) = 2x_4(-t) + \Omega_x(x_1(-t), x_2(-t)) = 2X_4(t) + \Omega_x(X_1(t), X_2(t)),\\ X_4'(t) &= -x_4'(-t) = 2x_3(-t) - \Omega_y(x_1(-t), x_2(-t)) = -2X_3(t) + \Omega_y(X_1(t), X_2(t)),\end{aligned}$$

since expressions (2) satisfy $\Omega_x(x,y) = \Omega_x(x,-y)$ and $\Omega_y(x,y) = -\Omega_y(x,-y)$.

We now present some basic concepts regarding the equilibrium points of the system and their stability. The computation of the equilibrium points and their characteristic eigenvalues is included in the *SageMath* code in the Appendix.

The system (1) has a total of 5 equilibrium points, which we denote by $L_i = L_i(\mu)$ for $i = 1,2,3,4,5$.
The points L_1, L_2, and L_3 lie on the x-axis with zero velocity. For all values of μ, they satisfy $x(L_2) < x(L_1) < 0 < x(L_3)$, where $x(L_i)$ denotes the first coordinate of L_i. These equilibrium points are computed numerically for each value of μ.
The points L_4 and L_5 share the same x-component, $x(L_4) = x(L_5)$, and have opposite non-zero y-components, $y(L_4) = -y(L_5)$, with zero velocity. The y-component of these equilibrium points is a constant that does not depend on μ,

and the x-component can be expressed as a function of μ.

Concerning stability, the linearized system at the equilibrium points $\{L_i\}_{i=1,2,3}$ is a center–saddle. By the theorem of invariant manifolds for flows, each equilibrium point L_i has associated one-dimensional stable and unstable invariant manifolds, which are smooth and tangent at L_i to the eigenvector of the linearized system evaluated at that point.

Part 2: Homoclinic orbits of L_3

In this section, we explain how to find homoclinic orbits of the equilibrium point L_3 by computing its stable and unstable invariant curves for different values of μ.

First, we recall that we can numerically obtain the stable and unstable curves associated to L_3, denoted by $W^s = W^s(L_3)$ and $W^u = W^u(L_3)$ respectively, following the procedure:

1. Let $v_{s,u}$ be the normalized eigenvector generating the stable (unstable) eigenspace of the Jacobian at L_3. We take $v_{s,u}$ with a non-negative first component.

2. We consider the points $L_3 \pm s \cdot v_{s,u}$ for some small value $s > 0$ (e.g: $s = 10^{-6}$).

3. We integrate the system backward (forward) in time to obtain the orbits of the two considered points. We denote by $W^{s,+}$ ($W^{u,+}$) the orbit of $L_3 + s \cdot v_{s,u}$, and by $W^{s,-}$ ($W^{u,-}$) the orbit of $L_3 - s \cdot v_{s,u}$.

4. We obtain the stable (unstable) curve as the union of the equilibrium point L_3 with the two computed orbits, that is, $W^s = L_3 \sqcup W^{s,+} \sqcup W^{s,-}$ ($W^u = L_3 \sqcup W^{u,+} \sqcup W^{u,-}$).

From now on, we focus on the "positive" branches $W^{s,+}$ and $W^{u,+}$. We note that by the symmetry of the system (1), the projections onto the (x, y)-plane of both branches are symmetric with respect to the x-axis. This implies that, if we consider the branches up to their first intersection with the x-axis—denoted by $W^{s,+}_{\text{cross}}$ and $W^{u,+}_{\text{cross}}$—their projection onto the (x, y)-plane, together with L_3, forms a closed curve. An example of such a curve is shown in Figure 1.

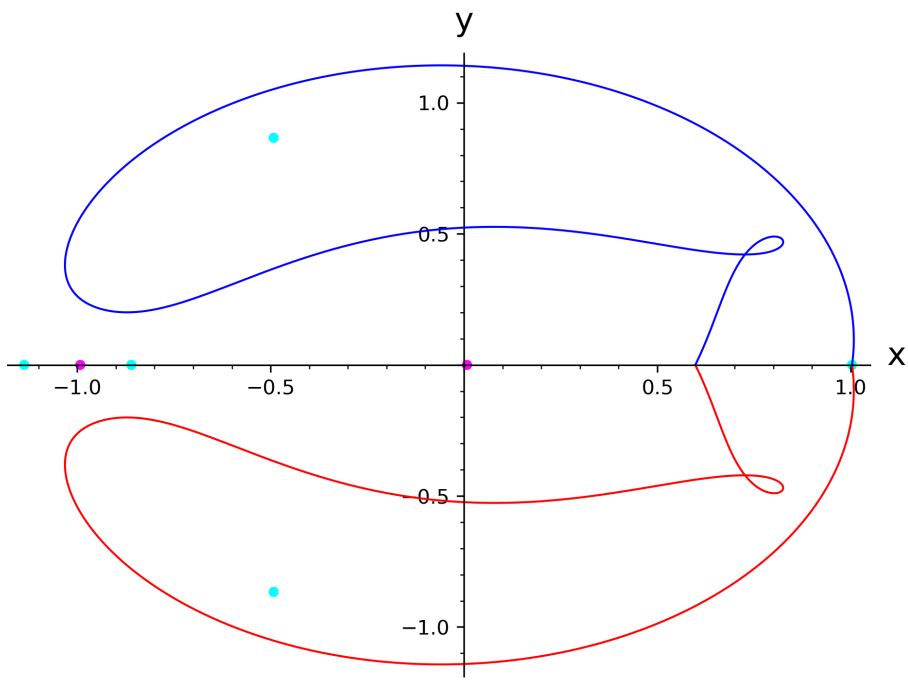

Figure 1: Projection onto the (x, y)-plane of $W^{s,+}(L_3)$ (blue) and $W^{u,+}(L_3)$ (red) up to their first intersection with the x-axis, for $\mu = 0.008$. The magenta dots indicate the positions of the primaries, P_1 and P_2, and the cyan dots indicate the equilibrium points of the system, L_i for $i = 1, 2, 3, 4, 5$.

We note that if $x_3 = x' = 0$ at the intersection of both branches with the x-axis, then $W^{s,+}_{\text{cross}} \sqcup W^{u,+}_{\text{cross}}$ is a homoclinic orbit of L_3. Therefore, by computing the intersection of the positive branches with the x-axis for different values of μ, one can examine the x_3-component at the intersection and determine whether $W^{s,+}_{\text{cross}} \sqcup W^{u,+}_{\text{cross}}$ forms a homoclinic orbit.

Part 3: Computation of the Positive Branches for Different μ

In this section, we present the results obtained using the *SageMath* code to compute $W^{u,+}_{\text{cross}}(L_3)$ and its symmetric curve $W^{s,+}_{\text{cross}}(L_3)$ for different mass parameters μ.

More precisely, for a partition of the mass-parameter interval $\mu \in (0, 0.5]$, we numerically compute the unstable and stable curves associated with L_3 up to their first intersection with the Poincaré section $y = 0$ (the x-axis). For each value of μ, we obtain the coordinates $x_3 = x'$ and $x_1 = x$ at the first intersection of $W^{u,+}(L_3)$ with the x-axis, and we analyse the corresponding plots

$x'(\mu)$ and $x(\mu)$. In particular, for those values of μ for which $x'(\mu) = 0$, the set $W^{s,+}_{\text{cross}}(L_3) \sqcup W^{u,+}_{\text{cross}}(L_3)$ forms a symmetric homoclinic orbit of L_3.

We start by presenting the general plots $x'(\mu)$ and $x(\mu)$ of $W^{u,+}(L_3)$ at its first intersection with the x-axis. Next, we analyse the behaviour of the system for the relevant values of μ, including the plots of $W^{s,+}_{\text{cross}}(L_3) \sqcup W^{u,+}_{\text{cross}}(L_3)$.

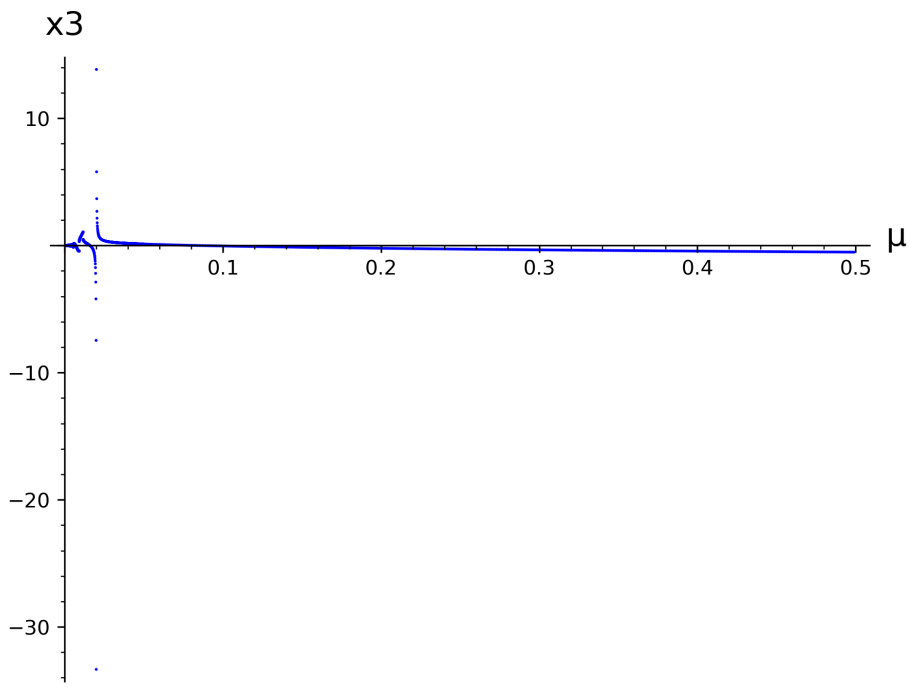

Figure 2: Plot of the crossing $x' = x_3$ of $W^{u,+}(L_3)$ at its first intersection with the x-axis, as a function of $\mu \in [0.001, 0.5]$. Remarkably, we observe a divergence (infinite jump) of x_3 for a value $\mu = \mu^* \approx 0.02$. For $\mu > \mu^*$, the function $x_3(\mu)$ is monotonically decreasing and intersects $x_3 = 0$ a single time.

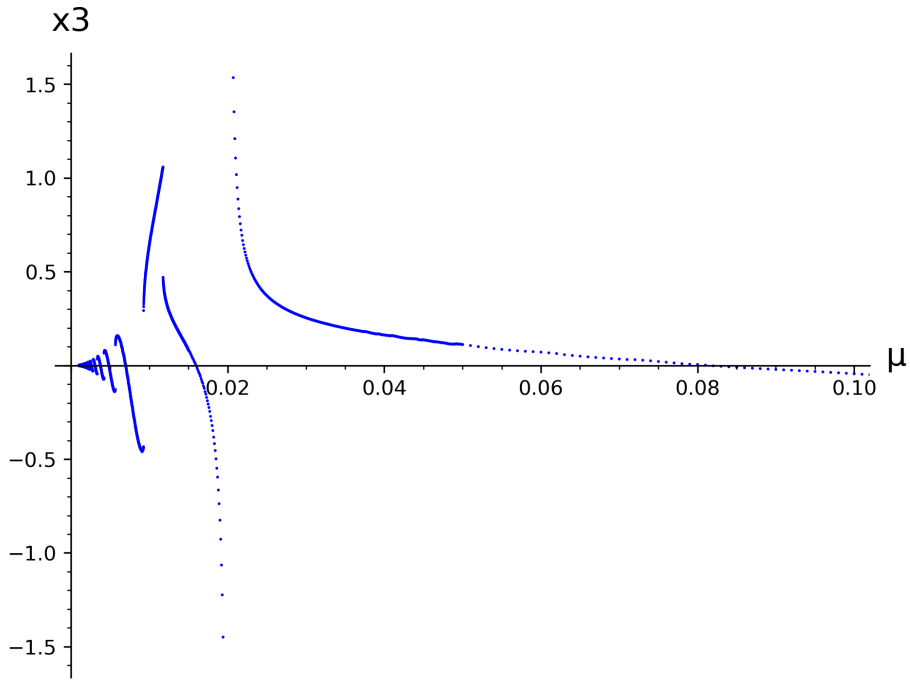

Figure 3: Plot of the crossing $x' = x_3$ of $W^{u,+}(L_3)$ at its first intersection with the x-axis, as a function of $\mu \in [0.001, 0.1]$. Besides the commented divergence of $x_3(\mu)$ for a value $\mu = \mu^* \approx 0.02$ and its decreasing behaviour for $\mu > \mu^*$, we observe that for $\mu < \mu^*$ the graphic $x_3(\mu)$ exhibits different finite jump discontinuities and intersections with $x_3 = 0$.

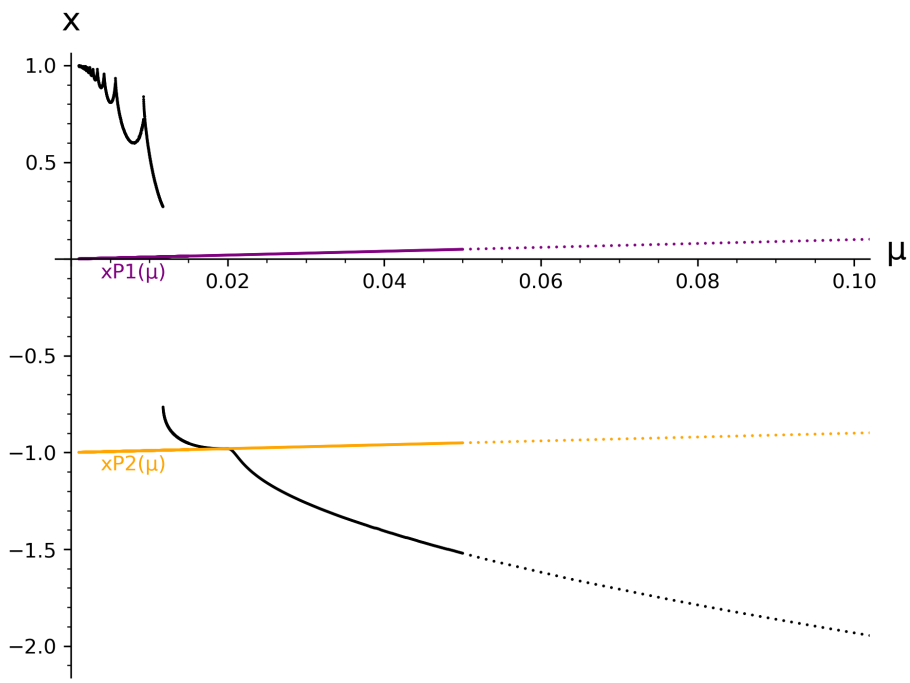

Figure 4: Plot of the crossing $x = x_1$ of $W^{u,+}(L_3)$ at its first intersection with the x-axis, as a function of $\mu \in [0.001, 0.1]$. We also plot the x-coordinates of the primaries $P_1(\mu)$ and $P_2(\mu)$. We observe that for $\mu \approx 0.012$ there is a finite jump discontinuity in $x(\mu)$, which in particular implies that $xP_1(\mu)$ is not a first intersection of $W^{u,+}(L_3)$ with the x-axis for any value of μ. Moreover, we observe that for μ in a neighbourhood of $\mu^* \approx 0.02$, we have $x(\mu) = xP_2(\mu)$.

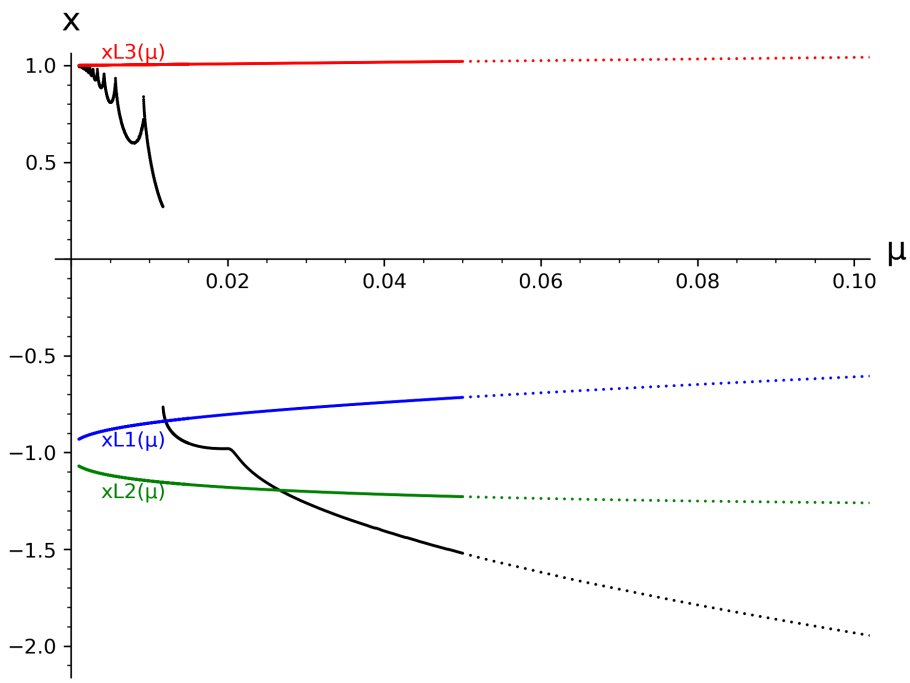

Figure 5: Plot of the crossing $x = x_1$ of $W^{u,+}(L_3)$ at its first intersection with the x-axis, as a function of $\mu \in [0.001, 0.1]$. We also plot the x-coordinates of the equilibrium points $L_i(\mu)$, $i = 1, 2, 3$. We note that $x(\mu)$ initially satisfies $x(\mu) \approx xL_3(\mu)$, and it also intersects $xL_1(\mu)$ and $xL_2(\mu)$ for two distinct values of μ. In particular, for the μ values at these intersections, the branch $W^{u,+}(L_3)$ forms a heteroclinic orbit from L_3 to L_1 and L_2.

Qualitative Analysis and Branch Plots

Let us now analyse in detail the behaviour of the system for relevant μ values. Precisely, we will qualitatively explain the discontinuities of the previous plots, and obtain the plot of the branches at the discontinuities and when they form homoclinic or heteroclinic orbits.

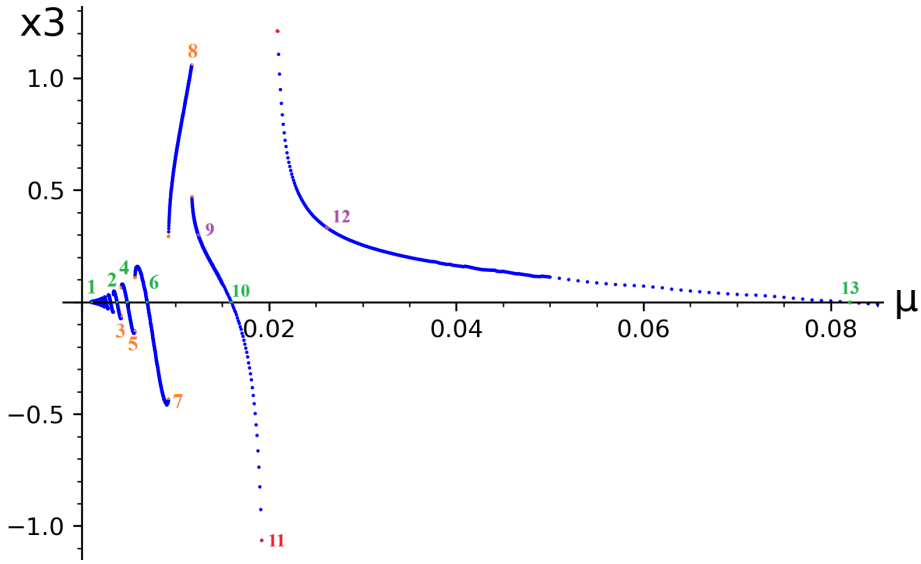

Figure 6: Same plot as the one described in Figure 3, but with some reference indications on relevant discontinuities (orange indications), homoclinic orbits (green indications) or heteroclinic orbits (purple indications).

For $\mu < \mu^* \approx 0.02$

For this parameter interval, the behaviour of the system is similar, although some relevant differences arise in certain cases. In particular, we observe a significant sensitivity of the invariant curves associated with L_3 to the value of μ.

We observe that for μ close to 0.001, the x' at the intersection is almost zero, and exhibits discontinuities jumping from negative to positive x'. We also observe that the curve $x'(\mu)$ crosses $x' = 0$ several times, which correspond to the homoclinic orbits we described earlier. Figures 7,8,11 and 14 are examples of these homoclinic orbits, which correspond to the green indications 1,2,4 and 6 on Figure 6, respectively.

The discontinuities in x' going from a negative value to a positive one are of the type represented in figures 9-10, 12-13 and 15-16, which correspond to the orange indications 3,5 and 7 in Figure 6, respectively. The discontinuities are caused because we transition from cases in which the orbit does a loop before

crossing $y = 0$, such that x' is negative at crossing (Figures 9,12 and 15), to cases in which there is no loop and x' is positive at crossing (Figures 10,13 and 16).

We now show the referenced figures, ordered by increasing μ.

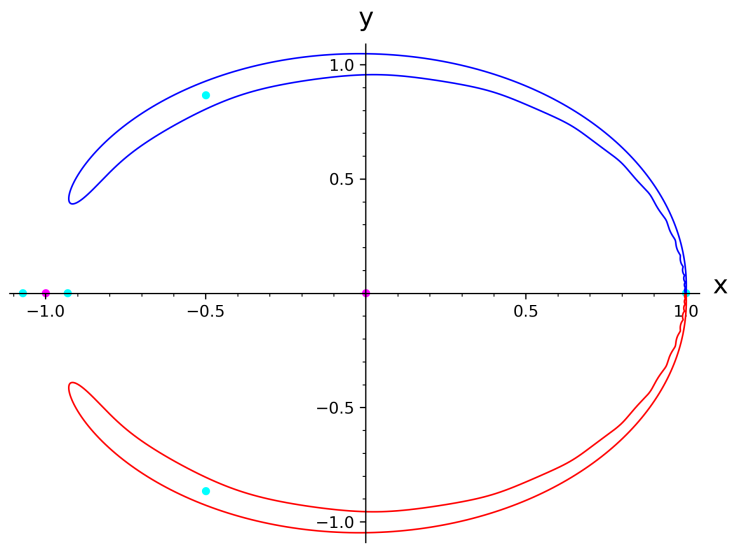

Figure 7: Homoclinic orbit, obtained for $\mu = 0.00103$. Corresponds to the indication 1 in Figure 6.

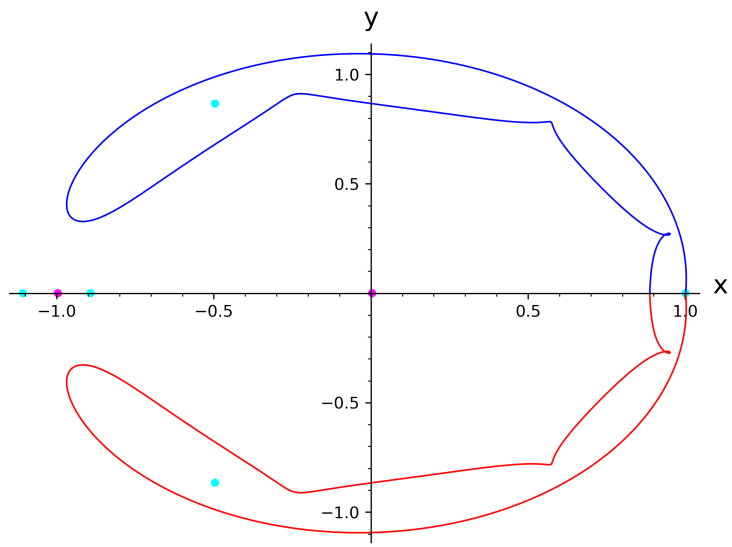

Figure 8: Homoclinic orbit, obtained for $\mu = 0.00374$. Corresponds to the indication 2 in Figure 6.

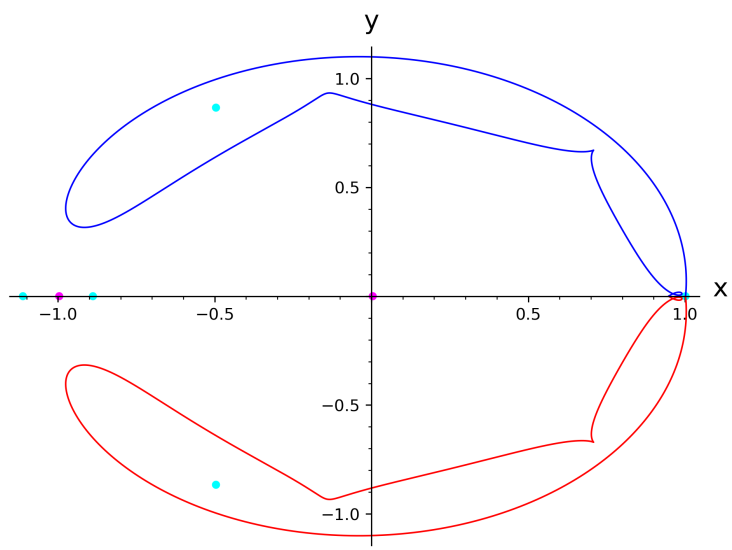

Figure 9: Positive branches, obtained for $\mu = 0.00418$. We observe a loop before crossing that causes x' to be negative at $y = 0$. Corresponds to the discontinuity 3 in Figure 6.

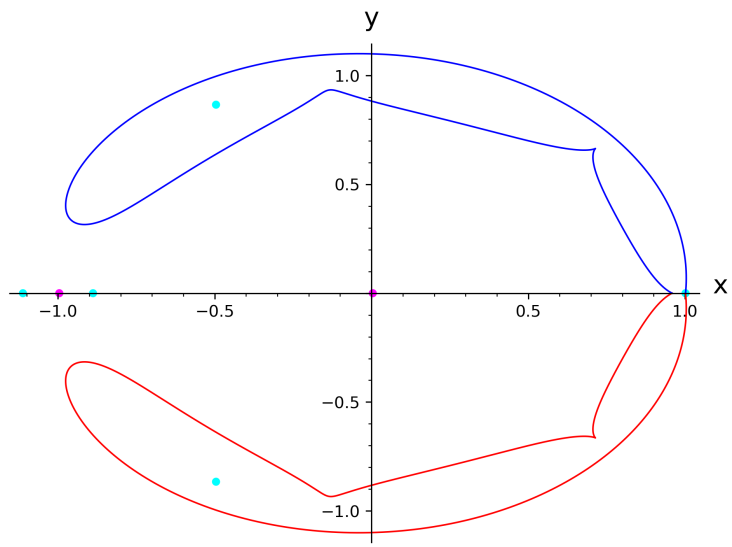

Figure 10: Positive branches, obtained for $\mu = 0.00420$. We observe that x' is positive at $y = 0$. Corresponds to the discontinuity 3 in Figure 6.

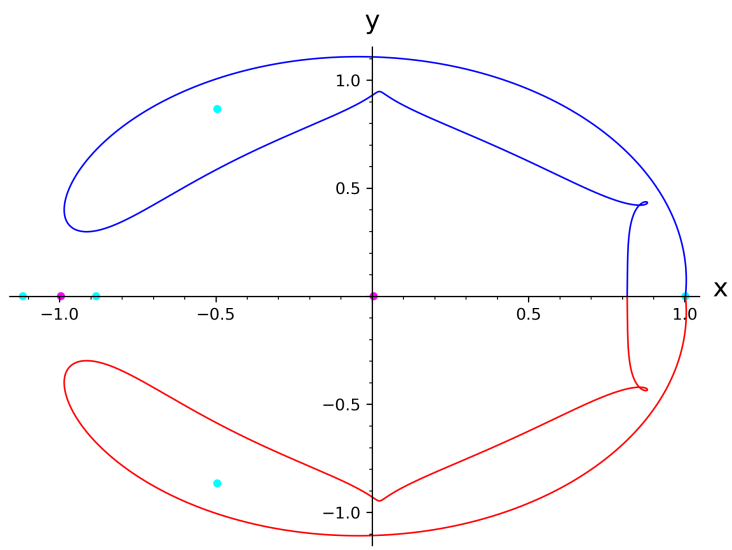

Figure 11: Homoclinic orbit, obtained for $\mu = 0.00483$. Corresponds to the indication 4 in Figure 6.

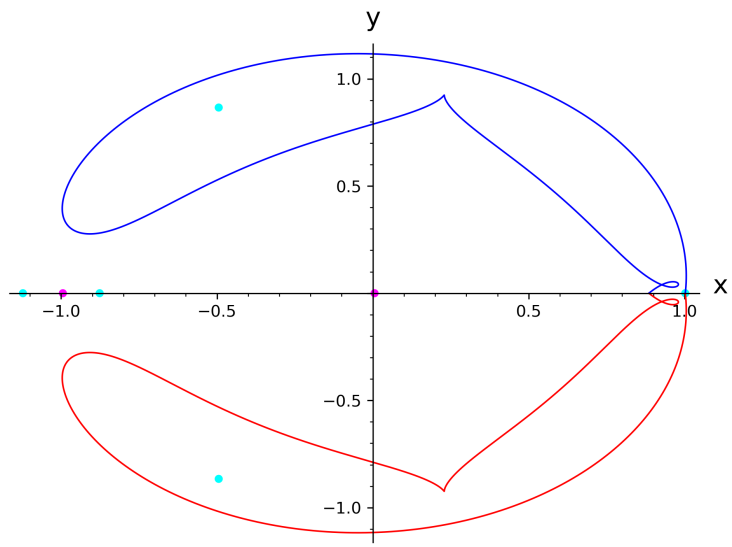

Figure 12: Positive branches, obtained for $\mu = 0.00560$. We observe a loop before crossing that causes x' to be negative at $y = 0$. Corresponds to the discontinuity 5 in Figure 6.

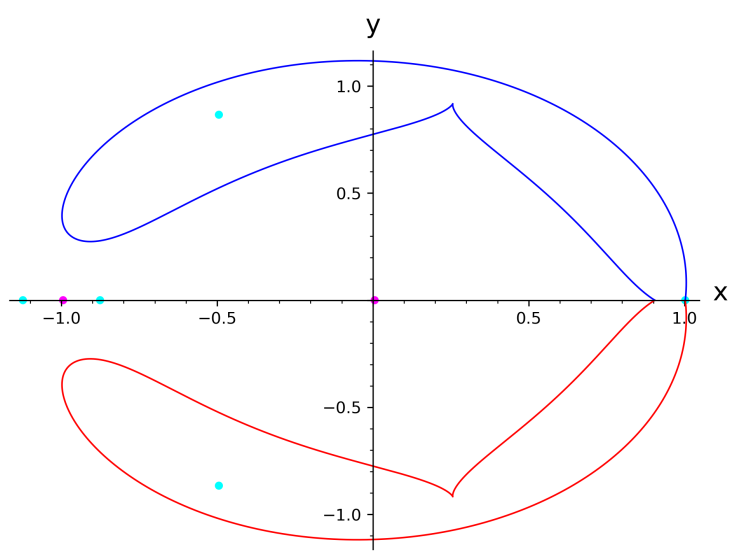

Figure 13: Positive branches, obtained for $\mu = 0.00570$. We observe that x' is positive at $y = 0$. Corresponds to the discontinuity 5 in Figure 6.

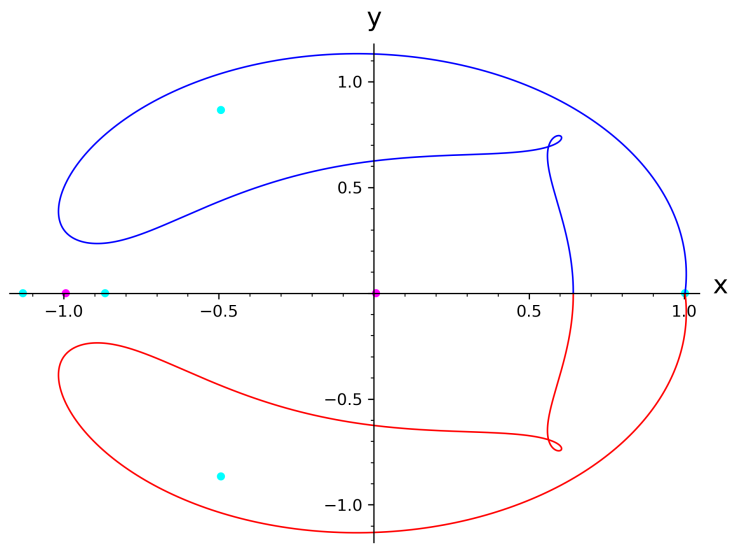

Figure 14: Homoclinic orbit, obtained for $\mu = 0.00698$. Corresponds to the indication 6 in Figure 6.

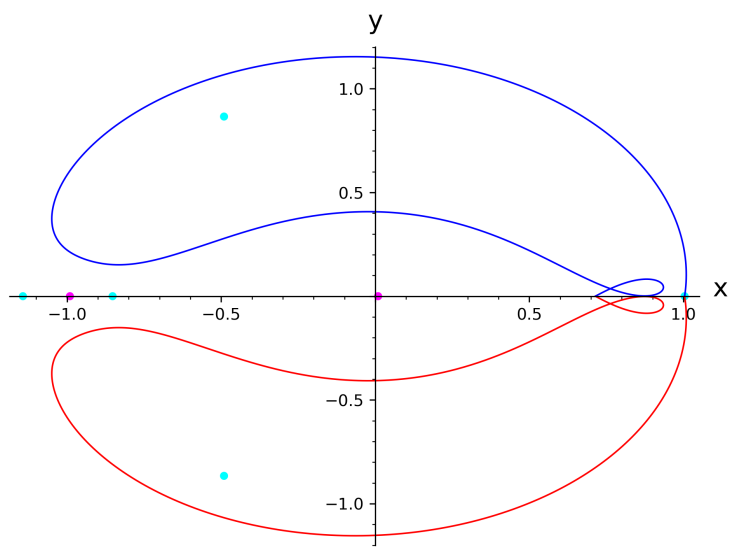

Figure 15: Positive branches, obtained for $\mu = 0.00925$. We observe a loop before crossing that causes x' to be negative at $y = 0$. Corresponds to the discontinuity 7 in Figure 6.

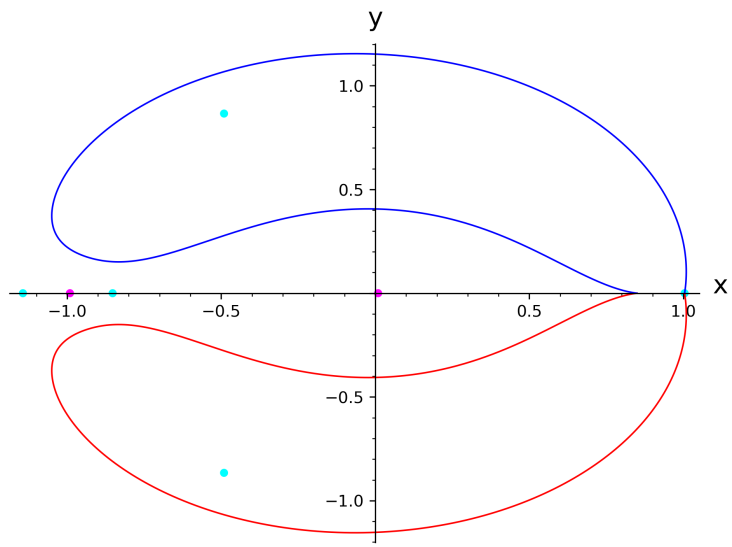

Figure 16: Positive branches, obtained for $\mu = 0.00926$. We observe that x' is positive at $y = 0$. Corresponds to the discontinuity 7 in Figure 6.

The following discontinuity, indicated as 8 in Figure 6, is different from the ones considered before. For this discontinuity, $x'(\mu)$ jumps from a positive x' to another positive x', but smaller. This discontinuity corresponds also to the $x(\mu)$ discontinuity observed in Figures 4 and 5, and implies that there is no impact between the first primary P_1 and a third body on the positive branch of the unstable orbit, $W^{u,+}(L_3)$, at least for a first intersection of $W^{u,+}(L_3)$ with the x-axis. The discontinuity is shown in Figures 17-18.

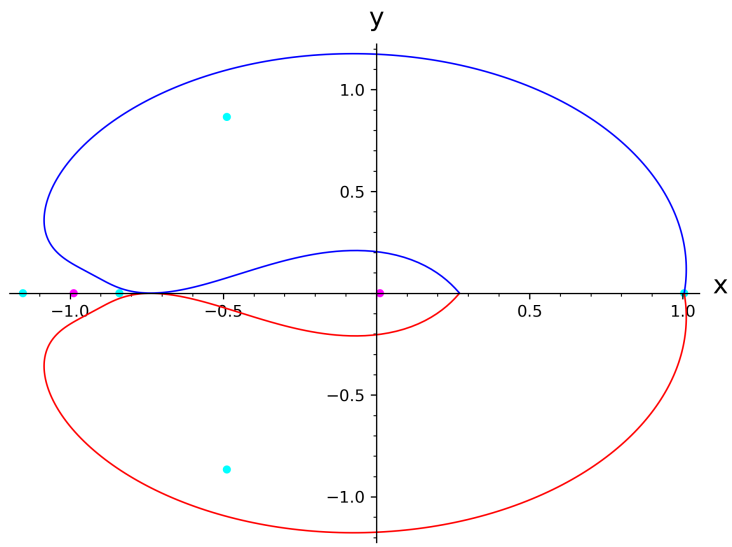

Figure 17: Positive branches, obtained for $\mu = 0.01173$. We observe that x' is positive at $y = 0$. Corresponds to the discontinuity 8 in Figure 6.

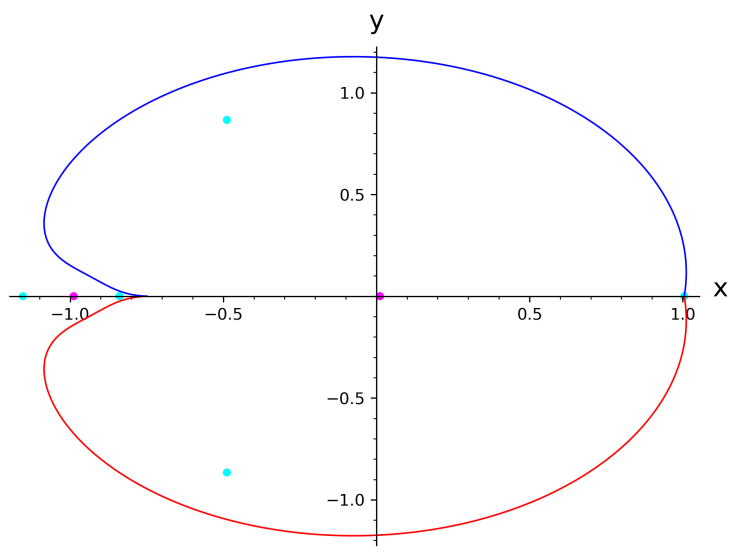

Figure 18: Positive branches, obtained for $\mu = 0.01174$. We observe that x' is positive at $y = 0$. With respect to the previous figure, there is an important discontinuity of x at the intersection, which is shown in Figure 4 and Figure 5. Corresponds to the discontinuity 8 in Figure 6.

Before the next homoclinic orbit, we indicate in Figure 6 a heteroclinic orbit

labelled as number 9. It corresponds to the intersection of the curve $x(\mu)$ with $xL_1(\mu)$ in Figure 5. Hence, it is a heteroclinic orbit from L_3 to L_1 for the unstable curve $W^u(L_3)$, and from L_1 to L_3 for the stable curve $W^s(L_3)$.

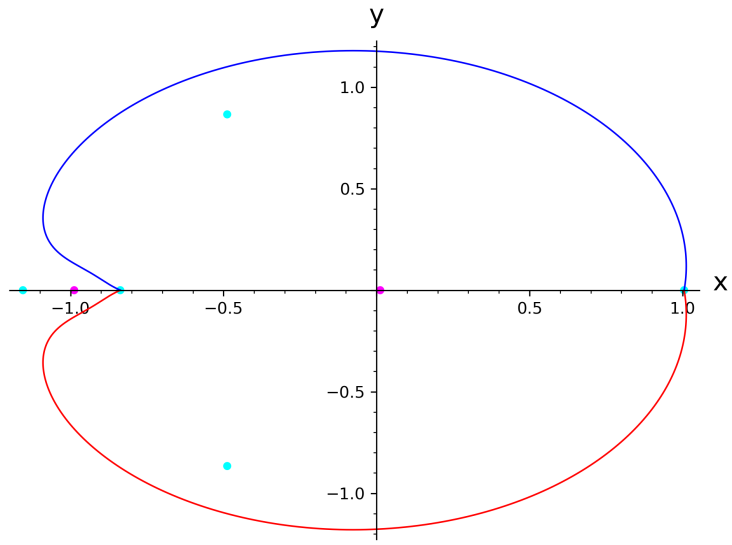

Figure 19: Heteroclinic orbits from L_3 to L_1 (red) and from L_1 to L_3 (blue), obtained for $\mu = 0.01208$. Corresponds to the indication 9 in Figure 6.

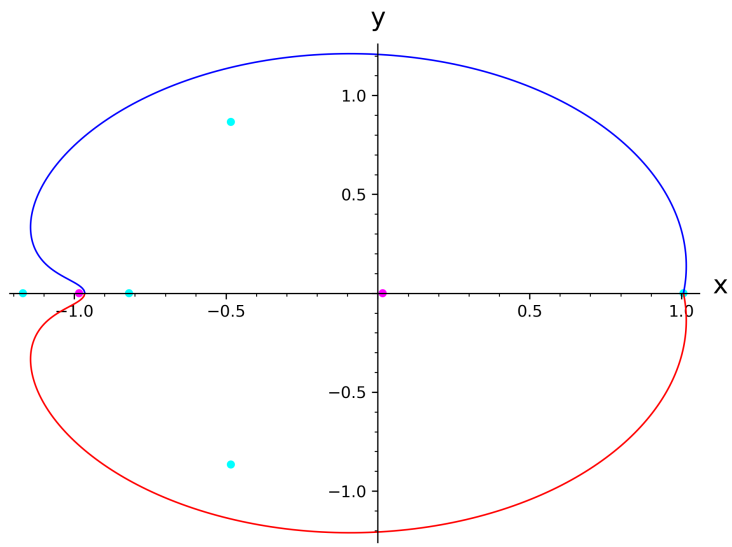

Figure 20: Homoclinic orbit, obtained for $\mu = 0.01600$. We note that this homoclinic orbit passes very close to the second primary P_2. Corresponds to the indication 10 in Figure 6.

Infinite Discontinuity for $\mu = \mu^* \approx 0.02$

For $\mu = \mu^* \approx 0.02$, we observe in Figure 6 an infinite discontinuity, labelled as 11. This feature corresponds either to a collision between the second primary P_2 and the third body along the unstable manifold associated with L_3, or to a trajectory of the third body from the second primary to the equilibrium point L_3 along the stable manifold associated with L_3.

To fix ideas, let us assume the first case (collision with the second primary); the other case is treated analogously. We observe that the x-component of the velocity of the third body, x', tends to infinity near the second primary, since the Newtonian gravitational force exerted by the primary on the third body increases proportionally to the inverse of their distance. The increase in $x' = x_3$ as the third body approaches the second primary can be seen directly from system (1), since Ω_x diverges as $r_2 \to 0$ (see expression (2)).
We also observe that $x'(\mu) \to -\infty$ as $\mu \to (\mu^*)^-$ and $x'(\mu) \to +\infty$ as $\mu \to (\mu^*)^+$. This behaviour can be explained as follows. For $\mu < \mu^*$, the crossing of the x-axis occurs at $x > xP_2$, so for $\mu \lesssim \mu^*$ the third body is attracted by the primary with a negative x' velocity. Analogously, for $\mu > \mu^*$, the crossing occurs at $x < xP_2$, so for $\mu \gtrsim \mu^*$ the third body is attracted by the primary with a positive x' velocity.

From Figure 4, we also observe that $x(\mu)$ at the intersection is equal to the

position of the second primary, xP_2, for μ in a neighbourhood of μ^*, which implies that a collision is possible for a range of μ values close to μ^*. In fact, μ^* represents a mathematical threshold; in a physical context, the primary has a non-zero spatial extension, which implies that collisions can occur for a finite range of μ values.

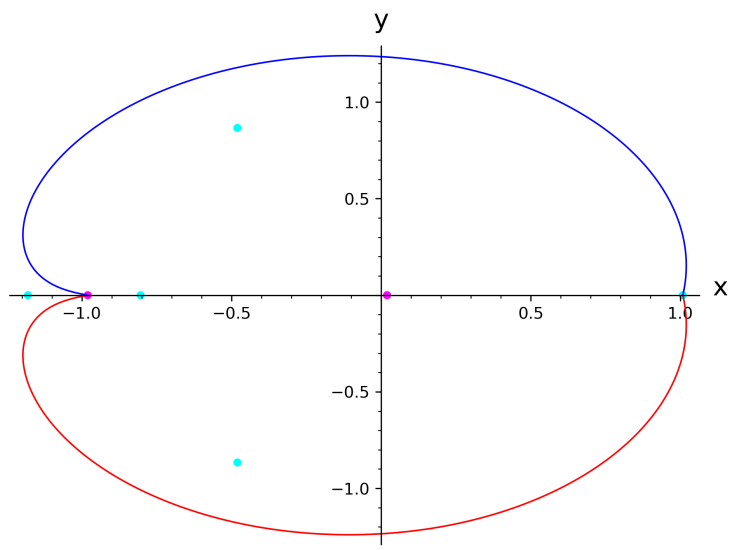

Figure 21: Positive branches, obtained for $\mu = 0.02000$, for the infinite discontinuity case described before. The same qualitative behaviour is obtained for μ in some neighbourhood of μ^*. Corresponds to the indication 11 in Figure 6.

For $\mu > \mu^* \approx 0.02$

For $\mu > \mu^*$, the graphic $x'(\mu)$ monotonically decreases as μ increases. Before crossing $x' = 0$ in the last homoclinic orbit (indication 13 in Figure 6), we have heteroclinic orbits from L_3 to L_2 for the unstable curve $W^u(L_3)$, and from L_2 to L_3 for the stable curve $W^s(L_3)$ (indication 12 in Figure 6).

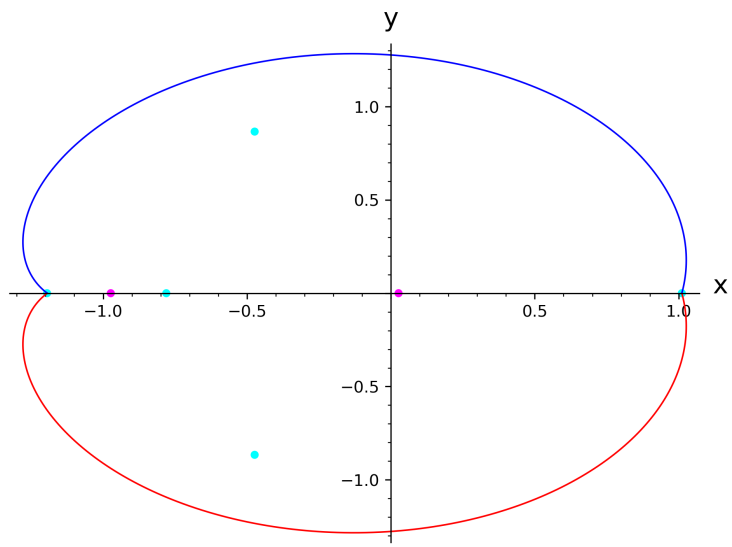

Figure 22: Heteroclinic orbits from L_3 to L_2 (red) and from L_2 to L_3 (blue), obtained for $\mu = 0.02660$. Corresponds to the indication 12 in Figure 6.

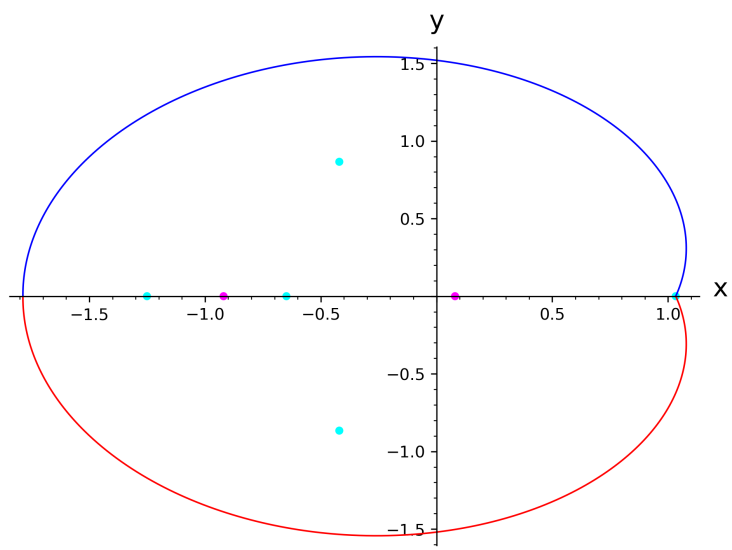

Figure 23: Homoclinic orbit, obtained for $\mu = 0.08000$. Corresponds to the indication 13 in Figure 6.

Appendix: *SageMath* Code

```
    #! /usr/bin/env sage
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import csv

from matplotlib.colors import Normalize
from scipy.integrate import solve_ivp
from mpl_toolkits.mplot3d import Axes3D


option=2




#SYSTEM AND INTEGRATION:

#POINCARÉ SECTION y=0

def g(t,x):
    return x[1]


def f(t,P,mu):
    x1=P[0]
    x2=P[1]
    x3=P[2]
    x4=P[3]
    o1=omegax(x1,x2,mu)
    o2=omegay(x1,x2,mu)
    return [x3,x4,2*x4+o1,-2*x3+o2]


def negf(t,P,mu):
    x1=P[0]
    x2=P[1]
    x3=P[2]
    x4=P[3]
    o1=omegax(x1,x2,mu)
    o2=omegay(x1,x2,mu)
    return [-x3,-x4,-2*x4-o1,2*x3-o2]
```

```
def L1(mu,TOL):
    def g(x,mu):
        a=mu*(1-x)**2
        b=3-2*mu-x*(3-mu-x)
        return (a/b)**(1/3)
    x0=(mu/(3-3*mu))**1/3
    while True:  #We do not write a break option because theoretically we know the numerical
        x1=g(x0,mu)
        if abs(x1-x0)<TOL:
            return [mu-1+x1,0,0,0]
        x0=x1
```

```
def L2(mu,TOL):
    def f(x,mu):
        a=mu*(1+x)**2
        b=3-2*mu+x*(3-mu+x)
        return (a/b)**(1/3)
    x0=(mu/(3-3*mu))**1/3
    while True:
        x1=f(x0,mu)
        if abs(x1-x0)<TOL:
            return [mu-1-x1,0,0,0]
        x0=x1
```

```
def L3(mu,TOL):
    def l(x,mu):
        a=(1-mu)*(1+x)**2
        b=1+2*mu+x*(2+mu+x)
        return (a/b)**(1/3)
    x0=1-7*mu/12
    while True:
        x1=l(x0,mu)
        if abs(x1-x0)<TOL:
            return [mu+x1,0,0,0]
        x0=x1
```

```
def L4(mu):
    return [mu-0.5,3**0.5/2,0,0]
```

```
def L5(mu):
    return [mu-0.5,-3**0.5/2,0,0]




def P1(mu):
    return [mu,0]

def P2(mu):
    return [mu-1,0]



#FUNCTIONS AND JACOBI 1st INTERGAL

def r1(x,y,mu):
    a=(x-mu)**2+y**2
    return sqrt(a)

def r2(x,y,mu):
    a=(x-mu+1)**2+y**2
    return sqrt(a)

def omega(x,y,mu):
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return (x**2+y**2)/2+(1-mu)/rr1+mu/rr2+mu*(1-mu)/2

def omegax(x,y,mu):
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return x-(1-mu)*(x-mu)/rr1**3-mu*(x-mu+1)/rr2**3

def omegay(x,y,mu):
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return y*(1-(1-mu)/rr1**3-mu/rr2**3)

def omegaxx(x,y,mu):
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return 1-(1-mu)/rr1**3+3*(1-mu)*(x-mu)**2/rr1**5-mu/rr2**3+3*mu*(x-mu+1)**2/rr2**5

def omegaxy(x,y,mu):
```

```
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return y*(3*(1-mu)*(x-mu)/rr1**5+3*mu*(x-mu+1)/rr2**5)

def omegayy(x,y,mu):
    rr1=r1(x,y,mu)
    rr2=r2(x,y,mu)
    return 1-(1-mu)/rr1**3-mu/rr2**3+y**2*(3*(1-mu)/rr1**5+3*mu/rr2**5)





#STABILITY:

def charexp(P,mu,TOL):
    x1=P[0]
    x2=P[1]
    x3=P[2]
    x4=P[3]
    oxx=omegaxx(x1,x2,mu)
    oxy=omegaxy(x1,x2,mu)
    oyy=omegayy(x1,x2,mu)
    A=np.array([[0,0,1,0],[0,0,0,1],[oxx,oxy,0,2],[oxy,oyy,-2,0]])
    evals,evecs=np.linalg.eig(A)

    cat1,cat2,cat3,cat4,cat5,cat6,cat7,cat8=[],[],[],[],[],[],[],[]
    for ev in evals:
        re=ev.real
        im=ev.imag
        if abs(im)<TOL:
            if re>TOL:
                cat1.append(ev)
            elif re<-TOL:
                cat2.append(ev)
        elif re>TOL:
            if im>TOL:
                cat3.append(ev)
            elif im<-TOL:
                cat4.append(ev)
        elif re<-TOL:
            if im>TOL:
                cat5.append(ev)
            elif im<-TOL:
                cat6.append(ev)
```

```
    elif abs(re)<TOL:
        if im>TOL:
            cat7.append(ev)
        elif im<-TOL:
            cat8.append(ev)
evals_sorted=np.array(cat1+cat2+cat3+cat4+cat5+cat6+cat7+cat8)


evecs_sorted=[]
for a in evals_sorted:
    i=np.where(np.isclose(evals,a))[0][0]
    v=evecs[:,i]

    v=v/np.linalg.norm(v)
    if v[0].real<0:
        v=-v

    evecs_sorted.append(v)
evals=[]
for a in evals_sorted:
    if abs(a.imag)<TOL:
        evals.append(a.real)
    elif abs(a.real)<TOL:
        evals.append(a.imag*1j)
    else:
        evals.append(a)

evals=np.array(evals)

evecs=[]
for v in evecs_sorted:
    v_aux=[]
    for a in v:
        if abs(a.imag)<TOL:
            v_aux.append(a.real)
        elif abs(a.real)<TOL:
            v_aux.append(a.imag)
        else:
            v_aux.append(a)
    evecs.append(v_aux)

evecs=np.array(evecs)

return A,evals,evecs
```

```
#MANIFOLD
#Type=0: Unstable manifold
#Type=1: Stable manifold
#Plot=0: Return mu, crossing x-velocity and crossing x
#Plot=1: Return solution for plotting



def manifold_crossing(pointindex,type,iregion,tf,mu,TOL,s,plot):

    if pointindex==1: P=L1(mu,TOL)
    elif pointindex==2: P=L2(mu,TOL)
    elif pointindex==3: P=L3(mu,TOL)
    else: return False

    v_standard=charexp(P,mu,TOL)[2][type]
    v=[]
    for a in v_standard:
        if iregion==1:
            v.append(a)
        elif iregion==-1:
            v.append(-a)
        else: return False

    v=np.array(v)


    x0=[P[0]+s*v[0],P[1]+s*v[1],P[2]+s*v[2],P[3]+s*v[3]]


    if type==0: #Forward in time
        f_fix=lambda t,x: f(t,x,mu)
        g.terminal = True
        if plot==1:
            solution=solve_ivp(f_fix,(0,tf),x0,max_step=0.01,atol=1e-12,events=g)
        if plot==0:
            solution=solve_ivp(f_fix,(0,tf),x0,atol=1e-12,events=g)

    if type==1: #Backward in time, assuming tf>0
        f_fix=lambda t,x: negf(t,x,mu)
        g.terminal = True
        if plot==1:
            solution=solve_ivp(f_fix,(0,tf),x0,max_step=0.01,atol=1e-12,events=g)
        if plot==0:
```

```
            solution=solve_ivp(f_fix,(0,tf),x0,atol=1e-12,events=g)


    times=solution.t
    points=solution.y


    crossingtime=solution.t_events[0][0]
    crossingpoint=solution.y_events[0][0]


    print("mu:")
    print(mu)
    print()
    print("crossing time:")
    print(crossingtime)
    print()
    print("Crossing x1:")
    print(crossingpoint[0])
    print()
    print("Crossing x3:")
    print(crossingpoint[2])
    print()
    print()

    if plot==0:
        return [mu,crossingpoint[2],crossingpoint[0]]
    if plot==1:
        return times,points



##########################################################################################


#MAIN:

mu=0.001
TOL=1e-12
s=1e-6
tf=1000
```

```
#What do we want?
#option=0: Plot for a given mu the positive branch (+1) of the unstable(type==0) and stable
#option=1: Write crossings as a function of mu of unstable manifold of L3
#option=2: Read crossings as a function of mu and plot

#Recall: manifold_crossing(pointindex,type,iregion,tf,mu,TOL,s,plot):









if option==0:

    plot=1
    plots=[]

    x_list_u=manifold_crossing(3,0,+1,tf,mu,TOL,s,plot)[1]
    plots.append(list_plot(list(zip(x_list_u[0],x_list_u[1])),color='red',plotjoined=True))

    x_list_s=manifold_crossing(3,1,+1,tf,mu,TOL,s,plot)[1]
    plots.append(list_plot(list(zip(x_list_s[0],x_list_s[1])),color='blue',plotjoined=True))

    primaries=[P1(mu),P2(mu)]
    plots.append(list_plot(primaries, color='magenta', size=25))

    equilibrium=[[L1(mu,TOL)[0],L1(mu,TOL)[1]],[L2(mu,TOL)[0],L2(mu,TOL)[1]],[L3(mu,TOL)[0],
    plots.append(list_plot(equilibrium, color='cyan', size=25))



    plot_total=sum(plots)
    #plot_total.set_axes_range(-1.5, 1.5, -1.2, 1.2)
    plot_total.axes_labels(['x','y'])
    plot_total.show(gridlines=True)
    plot_total.save(f'fA9-L3-mu={mu:.5f}.png',dpi=300)





if option==1:
    plot=0
```

```
    with open("data9.csv", "a", newline="") as file:
        writer=csv.writer(file)
        while mu<0.5:
            row=manifold_crossing(3,0,+1,tf,mu,TOL,s,plot)
            row.append(P1(mu)[0])
            row.append(P2(mu)[0])
            row.append(L1(mu,TOL)[0])
            row.append(L2(mu,TOL)[0])
            row.append(L3(mu,TOL)[0])

            writer.writerow(row)

            if mu<0.015:
                mu+=0.00001
            elif mu>=0.015 and mu<0.05:
                mu+=0.0001
            else:
                mu+=0.001




if option==2:
    df=pd.read_csv("data9.csv", header=None)
    print(df.shape)
    print(df.head())
    mu_vals  = df.iloc[:,int(0)].tolist()
    x3_vals  = df.iloc[:,int(1)].tolist()
    x1_vals  = df.iloc[:,int(2)].tolist()
    xP1_vals = df.iloc[:,int(3)].tolist()
    xP2_vals = df.iloc[:,int(4)].tolist()
    xL1_vals = df.iloc[:,int(5)].tolist()
    xL2_vals = df.iloc[:,int(6)].tolist()
    xL3_vals = df.iloc[:,int(7)].tolist()



    #Plot x'
    plots1=[]
    plots1.append(list_plot(list(zip(mu_vals, x3_vals)),size=2))

    plot1_total=sum(plots1)
    #plot1_total.set_axes_range(0, 0.1, -1.6, 1.6)
    plot1_total.axes_labels(['','x3'])
    plot1_total.show(gridlines=True)
    plot1_total.save(f'A9-L3-x3-mu.png',dpi=300)
```

```
#Plots x and primaries
plots2=[]
plots2.append(list_plot(list(zip(mu_vals, x1_vals)), size=2, color='black'))
plots2.append(list_plot(list(zip(mu_vals, xP1_vals)), size=2, color='purple'))
plots2.append(list_plot(list(zip(mu_vals, xP2_vals)), size=2, color='orange'))

mu_annot =0.008
labels2= [
    text('xP1()', (mu_annot, P1(mu_annot)[0]-0.08), color='purple', fontsize=10),
    text('xP2()', (mu_annot, P2(mu_annot)[0]-0.07), color='orange', fontsize=10),
]


plot2_total=sum(plots2)+sum(labels2)
plot2_total.set_axes_range(0, 0.1, -2.1, 1.0)
plot2_total.axes_labels(['','x'])
plot2_total.show(gridlines=True)
plot2_total.save(f'A9-L3-x-mu-primaries.png',dpi=300)



#Plots x and equilibrium points
plots3=[]
plots3.append(list_plot(list(zip(mu_vals, x1_vals)), size=2, color='black'))
plots3.append(list_plot(list(zip(mu_vals, xL1_vals)), size=2, color='blue'))
plots3.append(list_plot(list(zip(mu_vals, xL2_vals)), size=2, color='green'))
plots3.append(list_plot(list(zip(mu_vals, xL3_vals)), size=2, color='red'))

mu_annot = 0.008
labels3 = [
    text('xL1()', (mu_annot, L1(mu_annot,TOL)[0]-0.08), color='blue', fontsize=10),
    text('xL2()', (mu_annot, L2(mu_annot,TOL)[0]-0.07), color='green', fontsize=10),
    text('xL3()', (mu_annot, L3(mu_annot,TOL)[0]+0.06), color='red', fontsize=10)
]

plot3_total=sum(plots3)+sum(labels3)
plot3_total.set_axes_range(0, 0.1, -2.1, 1.0)
plot3_total.axes_labels(['','x'])
plot3_total.show(gridlines=True)
plot3_total.save(f'A9-L3-x-mu-lagrangians.png',dpi=300)
```